Import Data - ACF - PACF

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Setting R code chunk options

The first R code chunk is named **setup**. Here we are setting the options for R code chunks. The choice echo=TRUE means both code and output will appear on report. If for a specific chunk you want different options, you can always change that on the first line as we did in the following chunk. We chose include=FALSE which means that nothing related to this chunk (code and output) will appear on the knitted file.

Loading packages and initializing

It's useful to designate one code chunk to load packages on the beginning of the file. You can always add to this chunk as needed. But concentrate the packages needed on only one chunk.

Importing data

For this first example we will import water inflow data for reservoirs in Brazil. We have data for 15 different reservoirs spread all over the country. To import a .txt, .csv or .xlsx file you can use the function *read.table()*. This function will store the data as a data frame and has useful inputs such as

- *file =* : use this input to point to your data file. If it's on the same folder as your .Rmd then you only need to write the file name. But if it's on another folder you need to point to the path were file is located;
- *header =* : if your file has a header you should set this to TRUE, o.w. FALSE;
- *skip =* : if your file has rows explaining the data or any other rows on the top that need to be skipped you should just set skip to be equal to the number of row that should be skipped before reading the data. Note that if header=TRUE, you should not skip the row with the header. The default is *skip=0*;
- *dec =* : define *dec="."* or *dec=","* depending on how it's defined on your set. The default is ".".

```
#Importing time series data from text file
#I am calling it raw for now because it's data as it is in the file

#Option 1: import using absolute path
#raw_inflow_data <- read.table(file="/Users/lmm89/OneDrive/Duke_University/15_Spring2025/ENV797_TSA/Gi

#Option 2: import using shortcut "." or ".." or relative path
# the "." will point to the folder set as current directory
# the ".." will point to your parent folder set as current directory

#check which folder is your current directory using function getwd()
getwd()
```

```
## [1] "/home/guest/TSA_Sp26/Lessons"
```

```r
# my output was "/Users/lmm89/Library/CloudStorage/OneDrive-Personal/Duke_University/15_Spring2025/ENV7
# if output was "/Users/lmm89/Library/CloudStorage/OneDrive-Personal/Duke_University/15_Spring2025/ENV7

raw_inflow_data <- read.table(file="../Data/inflowtimeseries.txt", header=FALSE, skip=0)

#If you want specific columns just trim the table
raw_inflow_data <- raw_inflow_data[,1:5] #the space before the comma means you want all rows
                                  #and 1:5 means all columns from 1 to 5
nhydro <- ncol(raw_inflow_data)-2
nobs <- nrow(raw_inflow_data)

#If your file does not have header like this one you can add column names after creating the data frame
colnames(raw_inflow_data)=c("Month","Year", "HP1", "HP2","HP3")

#Checking data
head(raw_inflow_data)
```

```
##   Month Year  HP1  HP2  HP3
## 1   Jan 1931 4782 4076 2518
## 2   Feb 1931 7323 7681 4188
## 3   Mar 1931 8266 5921 3253
## 4   Apr 1931 6247 4600 2449
## 5   May 1931 3642 2789 1651
## 6   Jun 1931 2425 2062 1270
```

```r
str(raw_inflow_data)
```

```
## 'data.frame':    972 obs. of  5 variables:
##  $ Month: chr  "Jan" "Feb" "Mar" "Apr" ...
##  $ Year : int  1931 1931 1931 1931 1931 1931 1931 1931 1931 1931 ...
##  $ HP1  : int  4782 7323 8266 6247 3642 2425 2158 1854 1839 1896 ...
##  $ HP2  : int  4076 7681 5921 4600 2789 2062 1644 1301 1439 1340 ...
##  $ HP3  : int  2518 4188 3253 2449 1651 1270 1204 1152 1297 1259 ...
```

Date or time period

The data sets we will work with will be indexed by time, remember we are doing TIME series analysis. After importing your data set make sure that you have your dates right. For this specific inflow file our date is spread in two different columns. The first one is the month and the second the year. The best package to handle date conversion in R is lubridate. Let's see how we can use lubridate functions to combine those two columns into one date object. Note that this is only one example for our particular data set, for more info on lubridate function refer to tho this file file and "M1_Intro_DateObjects.Rmd" in our lessons folder.

```r
#using package lubridate
my_date <- paste(raw_inflow_data[,1],raw_inflow_data[,2],sep="-")
my_date <- my(my_date)  #function my from package lubridate
head(my_date)
```

```
## [1] "1931-01-01" "1931-02-01" "1931-03-01" "1931-04-01" "1931-05-01"
## [6] "1931-06-01"
```

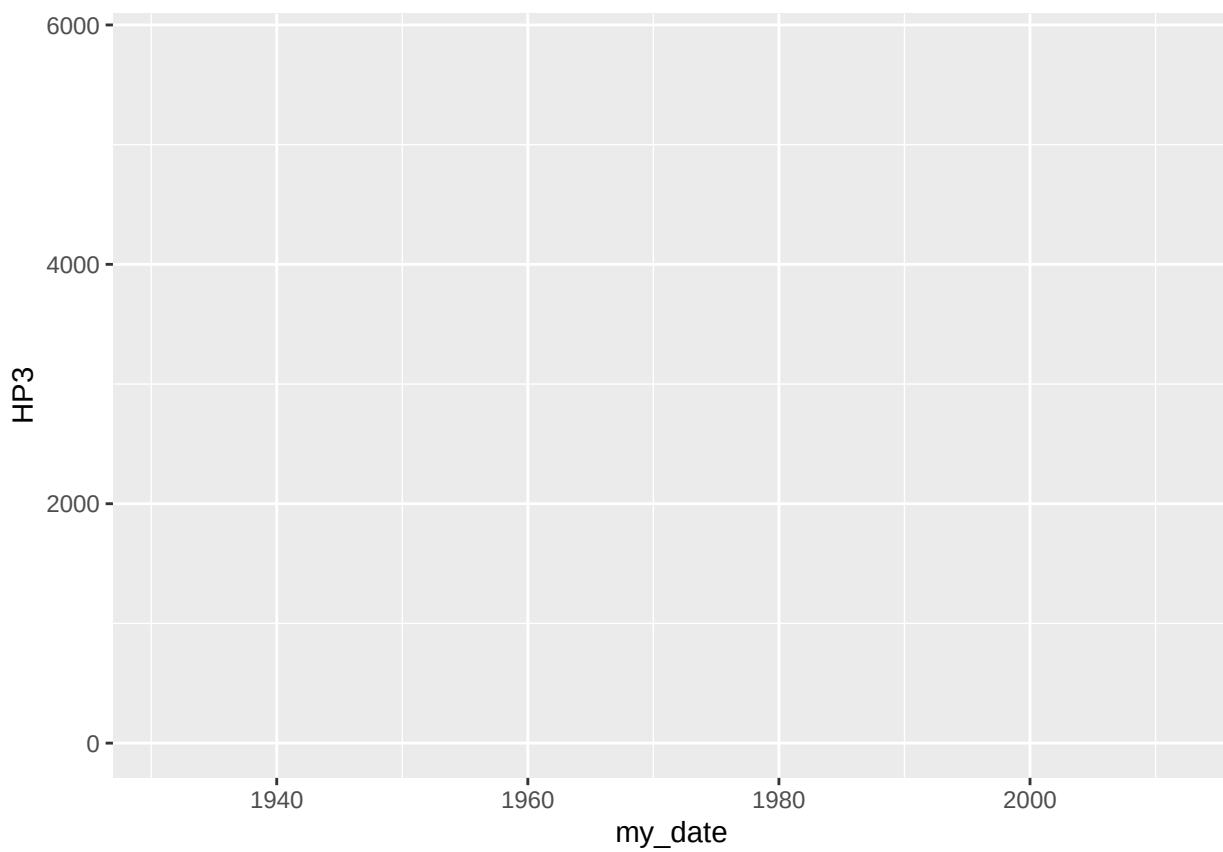
```r
#add that to inflow_data
inflow_data <- cbind(my_date,raw_inflow_data[,3:5]) #cbind stands for column bind
head(inflow_data)
```

```
##      my_date  HP1  HP2  HP3
## 1 1931-01-01 4782 4076 2518
## 2 1931-02-01 7323 7681 4188
## 3 1931-03-01 8266 5921 3253
## 4 1931-04-01 6247 4600 2449
## 5 1931-05-01 3642 2789 1651
## 6 1931-06-01 2425 2062 1270
```

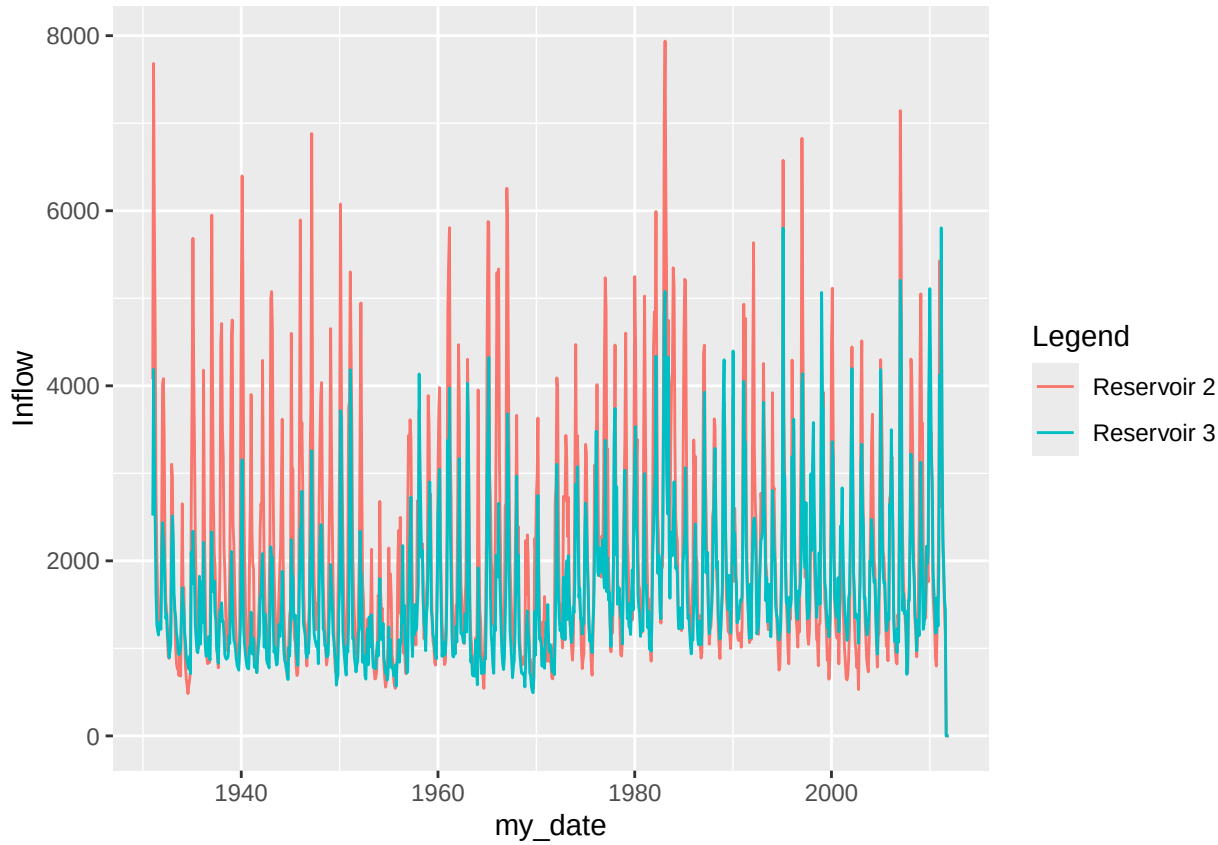
Initial plots

A plot of observed values over time is the first thing you do to start understanding the data set. The default packages on R offer the function *plot()*. But we can generate improved plots using another package *ggplot2*. Package ggplot2 offers better looking plots, additional functionalities, easy incorporation of the date labels, legends, etc.

```r
#using package ggplot2
ggplot(inflow_data, aes(x=my_date, y=HP3))
```



```r
#adding two time series to the same plot
ggplot(inflow_data, aes(x=my_date)) +
  geom_line(aes(y=HP2, color="Reservoir 2")) +
  geom_line(aes(y=HP3, color="Reservoir 3")) +
  labs(y="Inflow", color="Legend")
```



In this course we will also use *autoplot()* which is a ggplot() for time series object. But then we will need to convert our inflow data frame to a *ts* object first.

Transforming data into time series object

Many of the functions we will use require a time series object. You can transform your data in a time series using the function *ts()*. Only the columns with reservoir inflow data should be transformed, not the ones with month and year. Your ts object is not a data frame, but like I said we will need it as a ts for some function we will explore in future scripts.

```r
ts_inflow_data <- ts(inflow_data[, 2:4],
                     start = c(1931, 1),
                     frequency = 12)
ts_inflow_data
```

```
##            HP1  HP2  HP3
## Jan 1931  4782 4076 2518
## Feb 1931  7323 7681 4188
```

```
## Mar 1931  8266 5921 3253
## Apr 1931  6247 4600 2449
## May 1931  3642 2789 1651
## Jun 1931  2425 2062 1270
## Jul 1931  2158 1644 1204
## Aug 1931  1854 1301 1152
## Sep 1931  1839 1439 1297
## Oct 1931  1896 1340 1259
## Nov 1931  2095 1447 1218
## Dec 1931  2725 2479 2013
## Jan 1932  4679 4021 2435
## Feb 1932  5535 4082 2262
## Mar 1932  4310 3398 2065
## Apr 1932  3026 1965 1574
## May 1932  2185 1528 1341
## Jun 1932  1919 1431 1387
## Jul 1932  1640 1092 1146
## Aug 1932  1302  905 1095
## Sep 1932  1118  885  895
## Oct 1932  1688 1193  968
## Nov 1932  2040 1286 1049
## Dec 1932  3790 3104 2104
## Jan 1933  6928 2965 2513
## Feb 1933  5793 1641 2239
## Mar 1933  4276 1583 1634
## Apr 1933  3863 1381 1439
## May 1933  2498 1022 1354
## Jun 1933  1940  827 1178
## Jul 1933  1725  767 1109
## Aug 1933  1375  757  965
## Sep 1933  1324  690  925
## Oct 1933  1551  943  956
## Nov 1933  1724  684 1003
## Dec 1933  3352 1494 1341
## Jan 1934  4049 2652 1692
## Feb 1934  3166 1589 1541
## Mar 1934  3124 1694 1421
## Apr 1934  2507 1115 1183
## May 1934  1853  788 1065
## Jun 1934  1131  646  904
## Jul 1934   978  570  846
## Aug 1934   826  484  805
## Sep 1934  1026  574  764
## Oct 1934  1203  639  912
## Nov 1934  1199  743  714
## Dec 1934  1621 2642 2101
## Jan 1935  5047 3873 1744
## Feb 1935  5918 5685 2339
## Mar 1935  5340 4189 1952
## Apr 1935  5663 3286 1748
## May 1935  3407 2247 1260
## Jun 1935  2150 1628 1103
## Jul 1935  1610 1250  966
## Aug 1935  1430 1133  950
```

```
## Sep 1935   935 1006 1066
## Oct 1935  1305 1774 1824
## Nov 1935  1606 1383 1047
## Dec 1935  2458 1726 1224
## Jan 1936  2503 1691 1758
## Feb 1936  1804 1372 1299
## Mar 1936  4394 4179 2212
## Apr 1936  3197 2481 1399
## May 1936  2079 1681 1054
## Jun 1936  1424 1114  911
## Jul 1936  1269  900  920
## Aug 1936  1001  825  936
## Sep 1936   889 1059 1134
## Oct 1936   963  836  901
## Nov 1936  1317 1191  868
## Dec 1936  2120 2835 1714
## Jan 1937  3318 5950 2331
## Feb 1937  1642 3831 1741
## Mar 1937  2374 2695 1648
## Apr 1937  2600 2104 1767
## May 1937  1764 1927 1405
## Jun 1937  1556 1547 1322
## Jul 1937  1186 1098  995
## Aug 1937   943  889  926
## Sep 1937   878  778  830
## Oct 1937  1153 1401 1184
## Nov 1937  2243 1916 1428
## Dec 1937  3706 4496 1443
## Jan 1938  4379 4711 1520
## Feb 1938  3323 3497 1303
## Mar 1938  3227 3199 1312
## Apr 1938  2519 2144 1146
## May 1938  1617 1872  926
## Jun 1938  1359 1428  889
## Jul 1938  1259 1083  876
## Aug 1938   947  986  888
## Sep 1938   824  964  908
## Oct 1938  1109 1330 1141
## Nov 1938  1375 1945 1056
## Dec 1938  3173 3192 1601
## Jan 1939  4579 4563 2107
## Feb 1939  4886 4752 1716
## Mar 1939  2400 2422 1291
## Apr 1939  1995 2175 1092
## May 1939  1671 1682 1066
## Jun 1939  1459 1346  983
## Jul 1939  1086 1083  877
## Aug 1939   961  928  805
## Sep 1939   816  798  777
## Oct 1939   906  828  751
## Nov 1939  1623 1151 1020
## Dec 1939  1963 2743 1261
## Jan 1940  3368 4912 2119
## Feb 1940  5323 6398 3155
```

```
## Mar 1940  5088 5174 1885
## Apr 1940  2698 2899 1203
## May 1940  2120 2064 1048
## Jun 1940  1560 1523  938
## Jul 1940  1233 1201  857
## Aug 1940   973  968  790
## Sep 1940   813  825  770
## Oct 1940  1043  934  766
## Nov 1940  2576 1962 1017
## Dec 1940  2355 2617  987
## Jan 1941  4262 3901 1415
## Feb 1941  2899 2357  944
## Mar 1941  2527 1983  935
## Apr 1941  2765 1901 1115
## May 1941  1540 1100  789
## Jun 1941  1250  952  777
## Jul 1941  1069  908  815
## Aug 1941   848  728  723
## Sep 1941   908 1033  899
## Oct 1941  1114 1294 1037
## Nov 1941  1617 1569 1154
## Dec 1941  2614 2343 1570
## Jan 1942  3825 2651 1616
## Feb 1942  3636 2606 1925
## Mar 1942  5900 4290 2084
## Apr 1942  3646 2865 1431
## May 1942  2151 1930 1011
## Jun 1942  2938 1509 1394
## Jul 1942  1916 1236 1261
## Aug 1942  1150  982  868
## Sep 1942  1117  854  824
## Oct 1942  1189  914  815
## Nov 1942  2117 1588  777
## Dec 1942  3126 2266 1387
## Jan 1943  7617 4996 2160
## Feb 1943  7080 5076 2028
## Mar 1943  6210 4638 2041
## Apr 1943  3490 2763 1304
## May 1943  2195 1873  957
## Jun 1943  1983 1525 1012
## Jul 1943  1628 1231  895
## Aug 1943  1205 1049  810
## Sep 1943  1156  916  821
## Oct 1943  1765 1294 1030
## Nov 1943  2873 1598 1274
## Dec 1943  2960 1911 1134
## Jan 1944  2594 2161 1306
## Feb 1944  4140 2932 1492
## Mar 1944  3992 3619 1881
## Apr 1944  2772 2088 1192
## May 1944  2111 1437  974
## Jun 1944  1460 1105  859
## Jul 1944  1252  923  832
## Aug 1944  1017  769  758
```

```
## Sep 1944   840  672  726
## Oct 1944   998  675  644
## Nov 1944  2165 1071 1012
## Dec 1944  2291 1368  911
## Jan 1945  3372 1459 1220
## Feb 1945  6319 4599 2241
## Mar 1945  5967 3079 1855
## Apr 1945  6503 3037 1814
## May 1945  3587 1302 1347
## Jun 1945  2344 1118 1318
## Jul 1945  1926 1070 1381
## Aug 1945  1478  783  970
## Sep 1945  1200  690  882
## Oct 1945  1625  719  806
## Nov 1945  2692 1321 1208
## Dec 1945  5877 3323 1771
## Jan 1946  7288 5895 2318
## Feb 1946  5275 3314 2450
## Mar 1946  6616 3579 2796
## Apr 1946  4252 2852 1620
## May 1946  2808 1739 1277
## Jun 1946  2094 1334 1213
## Jul 1946  1926 1265 1148
## Aug 1946  1521 1007  940
## Sep 1946  1345  799  893
## Oct 1946  1390 1066  956
## Nov 1946  1764 1286  943
## Dec 1946  2809 1465 1146
## Jan 1947  3897 3186 1944
## Feb 1947  4721 3805 2380
## Mar 1947  7986 6882 3258
## Apr 1947  5620 3766 1870
## May 1947  2869 2264 1406
## Jun 1947  2051 1740 1165
## Jul 1947  1687 1547 1102
## Aug 1947  1436 1212 1053
## Sep 1947  1269 1509 1017
## Oct 1947  1366 1387 1075
## Nov 1947  1277 1501  825
## Dec 1947  3382 2655 1617
## Jan 1948  4968 3388 2177
## Feb 1948  4728 3886 2414
## Mar 1948  5399 4037 2409
## Apr 1948  3876 2633 1834
## May 1948  1893 1727 1222
## Jun 1948  1946 1409 1196
## Jul 1948  1351 1129 1031
## Aug 1948  1183  997 1037
## Sep 1948  1028  811  903
## Oct 1948  1030  884  904
## Nov 1948  1171 1326  928
## Dec 1948  3670 2200 1274
## Jan 1949  4132 2937 1551
## Feb 1949  6425 4653 1959
```

```
## Mar 1949  5453 3371 1752
## Apr 1949  3246 2441 1510
## May 1949  2353 1745 1169
## Jun 1949  1903 1190 1086
## Jul 1949  1756  893  946
## Aug 1949  1480  724  873
## Sep 1949  1255  744  580
## Oct 1949  1636  768  656
## Nov 1949  1934 1587  689
## Dec 1949  2996 2053 1383
## Jan 1950  3247 3246 1715
## Feb 1950  4325 6077 3716
## Mar 1950  4066 4213 3644
## Apr 1950  3033 3066 2129
## May 1950  2067 2175 1183
## Jun 1950  1629 1575 1011
## Jul 1950  1387 1281  888
## Aug 1950  1144 1008  775
## Sep 1950   990  827  695
## Oct 1950  1435 1079 1025
## Nov 1950  3080 1931  958
## Dec 1950  3753 3756 1286
## Jan 1951  5205 3816 2626
## Feb 1951  5183 5301 4184
## Mar 1951  4787 4430 3701
## Apr 1951  3793 3461 1932
## May 1951  2486 2116 1110
## Jun 1951  1981 1670 1123
## Jul 1951  1655 1378  896
## Aug 1951  1423 1156  824
## Sep 1951  1187  935  731
## Oct 1951  1279  937  811
## Nov 1951  1352  877  849
## Dec 1951  2007 1527 1164
## Jan 1952  3147 2033 1234
## Feb 1952  4503 3546 2339
## Mar 1952  8539 4946 1901
## Apr 1952  3877 2989 1789
## May 1952  2528 1751  841
## Jun 1952  2025 1574  925
## Jul 1952  1670 1208  910
## Aug 1952  1391  936  684
## Sep 1952  1289  846  649
## Oct 1952  1264  902 1023
## Nov 1952  1878 1336 1226
## Dec 1952  2689 1327  812
## Jan 1953  2089 1270 1359
## Feb 1953  2006 1277 1313
## Mar 1953  3551 1440 1017
## Apr 1953  3498 2133 1385
## May 1953  2098 1094  975
## Jun 1953  1496  903  782
## Jul 1953  1250  778  781
## Aug 1953  1037  651  774
```

```
## Sep 1953  1080  657  766
## Oct 1953  1643  722  943
## Nov 1953  1772 1014 1067
## Dec 1953  3319 1605  912
## Jan 1954  2436 1533 1138
## Feb 1954  3976 2679 1797
## Mar 1954  2240 1661 1311
## Apr 1954  1863 1404 1101
## May 1954  1573 1459 1101
## Jun 1954  1183 1134 1286
## Jul 1954   924  811  890
## Aug 1954   754  664  817
## Sep 1954   663  557  717
## Oct 1954   647  616  750
## Nov 1954  1443  736  638
## Dec 1954  2487 1102  701
## Jan 1955  3374 2148 1245
## Feb 1955  2765 1468  787
## Mar 1955  2309 1851 1099
## Apr 1955  2787 1583  858
## May 1955  1504  928  776
## Jun 1955  1167  850  771
## Jul 1955   930  657  687
## Aug 1955   751  570  693
## Sep 1955   635  543  809
## Oct 1955  1162  614  567
## Nov 1955  1548  899  833
## Dec 1955  3528 1959  868
## Jan 1956  3614 2347 1098
## Feb 1956  2460 1408  842
## Mar 1956  3452 2497 1042
## Apr 1956  2171 1363 1060
## May 1956  2164 1366 1561
## Jun 1956  2212 1386 2175
## Jul 1956  1539  947 1299
## Aug 1956  1321 1090 1493
## Sep 1956  1106  829 1096
## Oct 1956  1068  710  926
## Nov 1956  2262  961  718
## Dec 1956  4630 2472  741
## Jan 1957  5651 3432 1523
## Feb 1957  6578 3276 2107
## Mar 1957  6355 3612 2279
## Apr 1957  5401 3411 2730
## May 1957  3656 2119 1124
## Jun 1957  2720 1551  907
## Jul 1957  2035 1338 1287
## Aug 1957  1639 1087 1157
## Sep 1957  1503 1382 1502
## Oct 1957  1352 1040 1205
## Nov 1957  1849 1652 1249
## Dec 1957  3357 2686 1516
## Jan 1958  3849 2057 1726
## Feb 1958  4484 3726 4135
```

```
## Mar 1958  3823 2978 3054
## Apr 1958  3075 2233 2604
## May 1958  2470 2114 2037
## Jun 1958  1827 2197 2117
## Jul 1958  1656 1540 1568
## Aug 1958  1279 1110 1302
## Sep 1958  1234 1277 1181
## Oct 1958  1514 1357  973
## Nov 1958  1305 1640 1399
## Dec 1958  1955 1783 1498
## Jan 1959  5199 3888 2056
## Feb 1959  3550 3568 2528
## Mar 1959  5184 2716 2902
## Apr 1959  3327 2761 2252
## May 1959  2033 1723 1429
## Jun 1959  1639 1284 1345
## Jul 1959  1301 1080 1145
## Aug 1959  1120 1013 1124
## Sep 1959   947  835  973
## Oct 1959  1127  809  923
## Nov 1959  2111 1107  881
## Dec 1959  2130 1730 1603
## Jan 1960  3811 3019 2515
## Feb 1960  4799 3739 2828
## Mar 1960  4939 3979 3045
## Apr 1960  3501 2196 1983
## May 1960  2615 1713 1396
## Jun 1960  1950 1269  908
## Jul 1960  1591 1160 1016
## Aug 1960  1099 1029 1070
## Sep 1960   912  815  922
## Oct 1960  1174  844  948
## Nov 1960  2350 1354 1176
## Dec 1960  3886 2389 1727
## Jan 1961  6265 4870 3378
## Feb 1961  7520 5411 2615
## Mar 1961  6615 5806 3975
## Apr 1961  3646 3248 2351
## May 1961  2988 2727 1843
## Jun 1961  2078 1778 1175
## Jul 1961  1610 1394  913
## Aug 1961  1297 1183  902
## Sep 1961  1061 1044  984
## Oct 1961  1022  962  936
## Nov 1961  1547 1489 1252
## Dec 1961  2373 1510 1072
## Jan 1962  5226 2633 1561
## Feb 1962  5537 4469 2343
## Mar 1962  5958 3851 3168
## Apr 1962  3384 2289 1665
## May 1962  2416 1723 1171
## Jun 1962  1875 1551 1171
## Jul 1962  1421 1227 1142
## Aug 1962  1179 1069 1070
```

```
## Sep 1962  1148 1055 1063
## Oct 1962  1453 1527 1393
## Nov 1962  1698 1755 1263
## Dec 1962  6200 2952 1186
## Jan 1963  5166 4304 4031
## Feb 1963  5032 3519 3473
## Mar 1963  2962 2565 2230
## Apr 1963  2260 1519 1216
## May 1963  1764 1158  844
## Jun 1963  1548 1042  917
## Jul 1963  1395  884  699
## Aug 1963  1234  768  685
## Sep 1963  1075  688  697
## Oct 1963  1005  730  682
## Nov 1963  1156 1118 1094
## Dec 1963  1013  784  823
## Jan 1964  3422 1595  584
## Feb 1964  5142 3953 1921
## Mar 1964  3092 2213 1496
## Apr 1964  2662 1446  925
## May 1964  1748 1306  887
## Jun 1964  1210  959  712
## Jul 1964  1041  900  826
## Aug 1964   847  666  838
## Sep 1964   689  543  717
## Oct 1964  1365 1066  898
## Nov 1964  2106 1399  878
## Dec 1964  3351 2262 1467
## Jan 1965  4644 4655 2908
## Feb 1965  5776 5877 3605
## Mar 1965  7337 5862 4323
## Apr 1965  4546 2972 2252
## May 1965  2938 2693 1628
## Jun 1965  2123 2004 1153
## Jul 1965  1743 1832 1007
## Aug 1965  1373 1429  799
## Sep 1965  1116 1261  715
## Oct 1965  1968 1798 1261
## Nov 1965  2753 2212 1203
## Dec 1965  4503 3523 2067
## Jan 1966  6327 5294 1890
## Feb 1966  8228 4276 1817
## Mar 1966  6284 5335 2658
## Apr 1966  4172 3153 1933
## May 1966  3131 2328 1611
## Jun 1966  2253 1618 1222
## Jul 1966  1849 1293  962
## Aug 1966  1604 1056  820
## Sep 1966  1570 1014  758
## Oct 1966  1943 1368  891
## Nov 1966  2609 2625 1328
## Dec 1966  3611 2658 1758
## Jan 1967  4755 6257 2547
## Feb 1967  5528 5744 3681
```

```
## Mar 1967  5078 4131 3085
## Apr 1967  3791 2714 1848
## May 1967  2606 1927 1203
## Jun 1967  2413 1613 1032
## Jul 1967  1612 1342  856
## Aug 1967  1331 1165  665
## Sep 1967  1220 1071  793
## Oct 1967  1192  960  861
## Nov 1967  2171 1880 1182
## Dec 1967  3929 2609 1802
## Jan 1968  4152 3663 2968
## Feb 1968  4935 2158 1635
## Mar 1968  5302 2391 2067
## Apr 1968  3215 1373 1088
## May 1968  2252 1067  925
## Jun 1968  1824  917  802
## Jul 1968  1541  817  718
## Aug 1968  1480  836  726
## Sep 1968  1341  816  699
## Oct 1968  1789  907  708
## Nov 1968  2381  850  560
## Dec 1968  3663 2231  839
## Jan 1969  2819 1621 1282
## Feb 1969  3023 2297 1429
## Mar 1969  2609 2002 1194
## Apr 1969  1882 1211 1057
## May 1969  1571  863  717
## Jun 1969  1142  868  690
## Jul 1969   972  731  561
## Aug 1969   818  711  515
## Sep 1969   716  541  493
## Oct 1969  1163  921  667
## Nov 1969  3118 2254 1421
## Dec 1969  3167 2121  923
## Jan 1970  5578 2731 1516
## Feb 1970  5640 2982 2387
## Mar 1970  5192 3632 2748
## Apr 1970  3207 1890 1560
## May 1970  2134 1330 1104
## Jun 1970  1613 1099 1123
## Jul 1970  1358 1061 1059
## Aug 1970  1109  861  796
## Sep 1970  1161 1303  996
## Oct 1970  1490 1173  970
## Nov 1970  1838 1592  773
## Dec 1970  1585 1354 1095
## Jan 1971  1502 1259 1302
## Feb 1971  1538  853 1172
## Mar 1971  2038 1191 1501
## Apr 1971  1627 1048 1070
## May 1971  1150  851  962
## Jun 1971   966  915 1043
## Jul 1971   774  816  997
## Aug 1971   694  659  748
```

```
## Sep 1971   711  652  727
## Oct 1971  1204 1037  870
## Nov 1971  2023 1070  700
## Dec 1971  4236 2675 1144
## Jan 1972  3013 2879 1901
## Feb 1972  3746 4091 3104
## Mar 1972  3557 3958 2153
## Apr 1972  2727 2345 1610
## May 1972  1689 1537 1327
## Jun 1972  1300 1292 1202
## Jul 1972  1198 1497 1283
## Aug 1972   973 1179 1125
## Sep 1972   852 1005 1113
## Oct 1972  1938 2736 1810
## Nov 1972  3683 2722 1433
## Dec 1972  4262 2691 1100
## Jan 1973  4791 3434 1874
## Feb 1973  4662 3175 2000
## Mar 1973  4691 2281 1330
## Apr 1973  5038 2727 2059
## May 1973  2763 1617 1611
## Jun 1973  2048 1287 1387
## Jul 1973  1666 1191 1155
## Aug 1973  1286 1005 1066
## Sep 1973  1122  866 1181
## Oct 1973  1781 1040 1468
## Nov 1973  3247 1708 1409
## Dec 1973  3592 2912 1769
## Jan 1974  4230 4471 2875
## Feb 1974  2927 2889 2433
## Mar 1974  5698 3433 3076
## Apr 1974  5138 3065 2309
## May 1974  3164 1829 1582
## Jun 1974  2290 1536 1686
## Jul 1974  1786 1346 1337
## Aug 1974  1517  977 1327
## Sep 1974  1165  769 1162
## Oct 1974  1614  878 1347
## Nov 1974  1415  935 1281
## Dec 1974  2699 2183 2189
## Jan 1975  3843 3331 2661
## Feb 1975  3240 3239 2445
## Mar 1975  2502 2397 2009
## Apr 1975  2989 1802 1771
## May 1975  1853 1255 1397
## Jun 1975  1340 1125 1082
## Jul 1975  1242 1105 1145
## Aug 1975   927  829 1076
## Sep 1975   727  693  950
## Oct 1975  1056 1096 1255
## Nov 1975  2160 1684 1479
## Dec 1975  2484 3091 2050
## Jan 1976  2545 2553 2514
## Feb 1976  3169 3277 3481
```

```
## Mar 1976  3373 4014 2474
## Apr 1976  2446 3262 1870
## May 1976  1835 2085 1830
## Jun 1976  1436 2125 2149
## Jul 1976  1159 1998 1872
## Aug 1976   942 1815 1871
## Sep 1976  1147 2286 2057
## Oct 1976  1479 2124 2242
## Nov 1976  2608 2804 1856
## Dec 1976  5415 4229 1691
## Jan 1977  5646 5234 3376
## Feb 1977  5255 4886 2380
## Mar 1977  3093 2908 1647
## Apr 1977  3332 3279 2037
## May 1977  2288 1969 1551
## Jun 1977  1897 1649 1643
## Jul 1977  1428 1255 1308
## Aug 1977  1083  961 1021
## Sep 1977  1253 1320 1514
## Oct 1977  1305 1178 1387
## Nov 1977  2241 1826 1995
## Dec 1977  3720 3307 3426
## Jan 1978  5961 4465 3741
## Feb 1978  3655 2715 2074
## Mar 1978  4748 2801 2847
## Apr 1978  3554 1788 1504
## May 1978  2521 1642 1611
## Jun 1978  2264 1674 1693
## Jul 1978  1851 1270 1534
## Aug 1978  1386  925 1361
## Sep 1978  1330  913 1384
## Oct 1978  1438 1118 1051
## Nov 1978  2595 2088 1760
## Dec 1978  4644 2783 2221
## Jan 1979  6936 3373 3037
## Feb 1979  8581 4602 2290
## Mar 1979  5619 3276 1792
## Apr 1979  3994 2215 1339
## May 1979  2841 2025 1519
## Jun 1979  2380 1611 1251
## Jul 1979  1983 1347 1239
## Aug 1979  1684 1326 1159
## Sep 1979  1883 1650 1511
## Oct 1979  1730 1324 1897
## Nov 1979  2453 1877 1437
## Dec 1979  3146 2927 2516
## Jan 1980  6259 5248 3127
## Feb 1980  8647 4493 3534
## Mar 1980  4973 3113 2482
## Apr 1980  4444 3390 2220
## May 1980  2958 2098 1607
## Jun 1980  2290 1726 1469
## Jul 1980  1962 1670 1402
## Aug 1980  1540 1213 1138
```

```
## Sep 1980  1513 1203 1279
## Oct 1980  1488 1183 1236
## Nov 1980  2219 1696 1201
## Dec 1980  4409 3552 2177
## Jan 1981  5534 5025 2996
## Feb 1981  3414 3541 1739
## Mar 1981  3673 2646 1724
## Apr 1981  3647 2029 1486
## May 1981  2281 1552 1235
## Jun 1981  1937 1565 1422
## Jul 1981  1574 1175  991
## Aug 1981  1306 1094 1082
## Sep 1981  1104  853  970
## Oct 1981  2011 1718 1300
## Nov 1981  4401 2715 2087
## Dec 1981  5704 4428 2336
## Jan 1982  8647 4846 2808
## Feb 1982  7456 4510 3000
## Mar 1982  9109 5991 4339
## Apr 1982  6422 3958 2557
## May 1982  4187 2554 1859
## Jun 1982  3320 2271 2072
## Jul 1982  2625 1899 1844
## Aug 1982  2207 1578 1411
## Sep 1982  1983 1288 1339
## Oct 1982  2511 1902 2039
## Nov 1982  2727 1941 2087
## Dec 1982  3909 3774 3311
## Jan 1983  8586 7005 4782
## Feb 1983 10915 7938 5078
## Mar 1983  8002 6244 4089
## Apr 1983  6424 5105 2986
## May 1983  4180 3651 2537
## Jun 1983  3445 4748 4328
## Jul 1983  2773 2995 2224
## Aug 1983  2220 2236 1573
## Sep 1983  2398 3379 2241
## Oct 1983  3004 4041 2324
## Nov 1983  3679 4154 2059
## Dec 1983  6514 5348 2407
## Jan 1984  5439 5124 2903
## Feb 1984  4029 3387 2352
## Mar 1984  3723 2508 1914
## Apr 1984  4383 2296 1941
## May 1984  2986 2236 1848
## Jun 1984  2072 1484 1341
## Jul 1984  1737 1238 1222
## Aug 1984  1768 1270 1379
## Sep 1984  1863 1357 1462
## Oct 1984  1690 1203 1295
## Nov 1984  1834 1247 1228
## Dec 1984  3417 3003 1970
## Jan 1985  6607 4552 2122
## Feb 1985  6135 5217 2323
```

```
## Mar 1985  6300 5198 3063
## Apr 1985  4465 3435 2314
## May 1985  2985 2262 1629
## Jun 1985  2309 1840 1342
## Jul 1985  1941 1509 1377
## Aug 1985  1661 1282 1198
## Sep 1985  1683 1191 1037
## Oct 1985  1692 1083  937
## Nov 1985  2136 1657 1124
## Dec 1985  2740 1791 1097
## Jan 1986  4773 3381 1557
## Feb 1986  4337 2947 2156
## Mar 1986  3809 3194 2420
## Apr 1986  2724 2064 1586
## May 1986  2233 1991 1372
## Jun 1986  1720 1357 1100
## Jul 1986  1525 1304 1041
## Aug 1986  1586 1375 1312
## Sep 1986  1228 1026 1126
## Oct 1986  1268  889 1042
## Nov 1986  1374  952 1167
## Dec 1986  3310 2605 2100
## Jan 1987  4267 4367 2567
## Feb 1987  4416 4464 3928
## Mar 1987  4684 3079 2603
## Apr 1987  3737 2630 1864
## May 1987  2634 2014 2070
## Jun 1987  1990 1716 2096
## Jul 1987  1573 1382 1414
## Aug 1987  1387 1048 1169
## Sep 1987  1342 1217 1264
## Oct 1987  1657 1252 1377
## Nov 1987  2397 1471 1520
## Dec 1987  5159 2501 1743
## Jan 1988  4722 2587 2316
## Feb 1988  5862 3622 2835
## Mar 1988  6350 3533 3284
## Apr 1988  4535 2442 2485
## May 1988  2970 1834 1916
## Jun 1988  2413 1663 1994
## Jul 1988  1842 1199 1382
## Aug 1988  1534 1014 1265
## Sep 1988  1391  884 1105
## Oct 1988  1896 1249 1213
## Nov 1988  2386 1465 1444
## Dec 1988  3405 1627 1743
## Jan 1989  4140 3694 3811
## Feb 1989  5188 3806 4297
## Mar 1989  4560 3500 3015
## Apr 1989  3072 2191 2277
## May 1989  2227 1569 1837
## Jun 1989  1865 1416 1671
## Jul 1989  1697 1199 1442
## Aug 1989  1546 1178 1835
```

```
## Sep 1989  1587 1155 1469
## Oct 1989  1388  999 1178
## Nov 1989  2467 1269 1616
## Dec 1989  6797 2528 2203
## Jan 1990  6130 3877 4398
## Feb 1990  3796 1768 1966
## Mar 1990  3700 2599 2316
## Apr 1990  2815 2176 1835
## May 1990  2379 1659 1762
## Jun 1990  1752 1206 1293
## Jul 1990  1581 1111 1378
## Aug 1990  1396 1089 1335
## Sep 1990  1503 1111 1460
## Oct 1990  1667 1177 1552
## Nov 1990  1707 1013 1538
## Dec 1990  1923 1270 1596
## Jan 1991  3776 3497 2284
## Feb 1991  4794 4932 4050
## Mar 1991  6554 4121 3366
## Apr 1991  6675 4770 3360
## May 1991  3228 2915 2193
## Jun 1991  2416 1960 1773
## Jul 1991  1932 1616 1562
## Aug 1991  1621 1265 1315
## Sep 1991  1470 1066 1115
## Oct 1991  1958 1590 1734
## Nov 1991  2096 1099 1121
## Dec 1991  2975 1850 1849
## Jan 1992  5101 4094 2235
## Feb 1992  7705 5634 2462
## Mar 1992  5164 3274 2490
## Apr 1992  4805 2683 2322
## May 1992  3592 2431 1943
## Jun 1992  2389 1598 1167
## Jul 1992  1972 1433 1212
## Aug 1992  1653 1160 1256
## Sep 1992  1946 1577 1733
## Oct 1992  2539 2050 1962
## Nov 1992  4136 2773 2233
## Dec 1992  4909 2631 2261
## Jan 1993  3786 2434 2362
## Feb 1993  5860 4254 3812
## Mar 1993  4452 3589 2677
## Apr 1993  4016 3081 2311
## May 1993  2582 1969 1543
## Jun 1993  2277 1860 1807
## Jul 1993  1703 1349 1303
## Aug 1993  1650 1213 1440
## Sep 1993  1687 1243 1731
## Oct 1993  1724 1385 1727
## Nov 1993  1697 1171 1133
## Dec 1993  3832 1793 1713
## Jan 1994  7282 3920 2805
## Feb 1994  4702 2743 2519
```

```
## Mar 1994  7249 2834 2150
## Apr 1994  4467 2187 1982
## May 1994  3014 1839 1487
## Jun 1994  2411 1409 1392
## Jul 1994  2032 1179 1349
## Aug 1994  1726  989 1190
## Sep 1994  1338  752 1097
## Oct 1994  1428  786 1154
## Nov 1994  2232 1237 1243
## Dec 1994  3672 1854 1723
## Jan 1995  3888 2548 2412
## Feb 1995  6525 6576 5801
## Mar 1995  3936 3659 3044
## Apr 1995  3788 2978 2937
## May 1995  3077 2137 1915
## Jun 1995  2149 1544 1521
## Jul 1995  1737 1316 1576
## Aug 1995  1351  986 1333
## Sep 1995  1271  824 1186
## Oct 1995  1447 1360 1587
## Nov 1995  1831 1331 1473
## Dec 1995  2487 1930 1561
## Jan 1996  3615 4294 3192
## Feb 1996  2758 2795 2320
## Mar 1996  3612 3488 3619
## Apr 1996  2492 2330 2050
## May 1996  1852 1790 1750
## Jun 1996  1474 1350 1440
## Jul 1996  1292 1203 1339
## Aug 1996  1158 1014 1331
## Sep 1996  1254 1486 1662
## Oct 1996  1102 1188 1596
## Nov 1996  2665 2429 1727
## Dec 1996  3426 3424 2014
## Jan 1997  6828 6827 4028
## Feb 1997  4004 4711 4138
## Mar 1997  4450 3351 2394
## Apr 1997  4004 2348 2033
## May 1997  2856 1810 1918
## Jun 1997  2899 2291 2666
## Jul 1997  2037 1519 1563
## Aug 1997  1631 1140 1335
## Sep 1997  1459 1047 1357
## Oct 1997  1613 1171 1472
## Nov 1997  2133 1585 2228
## Dec 1997  4040 2863 2995
## Jan 1998  3574 2482 2120
## Feb 1998  4676 3104 3158
## Mar 1998  4073 3047 3580
## Apr 1998  2967 2030 2658
## May 1998  2334 1707 2154
## Jun 1998  1916 1500 1686
## Jul 1998  1523 1101 1353
## Aug 1998  1432 1087 1557
```

```
## Sep 1998  1120  802 1646
## Oct 1998  1361 1275 2092
## Nov 1998  2026 1284 1505
## Dec 1998  2921 2309 2662
## Jan 1999  3490 4200 5066
## Feb 1999  2694 3653 3948
## Mar 1999  4651 3929 3698
## Apr 1999  2458 2219 2243
## May 1999  1793 1548 1760
## Jun 1999  1506 1414 1663
## Jul 1999  1260 1192 1412
## Aug 1999  1120  861 1234
## Sep 1999  1194  944 1358
## Oct 1999  1043  648 1132
## Nov 1999  1691  873 1304
## Dec 1999  2192 1625 1746
## Jan 2000  4040 4431 2571
## Feb 2000  5519 5115 3361
## Mar 2000  6331 3674 3173
## Apr 2000  3565 2422 2118
## May 2000  2286 1531 1589
## Jun 2000  1797 1313 1491
## Jul 2000  1565 1155 1356
## Aug 2000  1418 1114 1412
## Sep 2000  1777 1483 1806
## Oct 2000  1167  823 1191
## Nov 2000  2491 1389 1564
## Dec 2000  4098 2158 2398
## Jan 2001  3376 2130 2304
## Feb 2001  2617 1836 2834
## Mar 2001  3304 1540 2230
## Apr 2001  2517 1311 1750
## May 2001  1952  994 1496
## Jun 2001  1566  837 1337
## Jul 2001  1234  667 1236
## Aug 2001  1067  640 1092
## Sep 2001  1097  691 1131
## Oct 2001  1352  890 1577
## Nov 2001  2139 1223 1627
## Dec 2001  3383 2004 2264
## Jan 2002  5463 3161 3248
## Feb 2002  6519 4444 4197
## Mar 2002  4598 3153 2956
## Apr 2002  2924 1809 1873
## May 2002  2212 1357 1685
## Jun 2002  1754 1073 1353
## Jul 2002  1485  967 1394
## Aug 2002  1191  782 1338
## Sep 2002  1227  894 1346
## Oct 2002   867  530 1058
## Nov 2002  1457 1196 1468
## Dec 2002  2168 2076 1760
## Jan 2003  4813 3851 3129
## Feb 2003  4502 4514 3333
```

```
## Mar 2003  4384 3021 2578
## Apr 2003  4049 2475 2380
## May 2003  2394 1714 1572
## Jun 2003  1856 1360 1585
## Jul 2003  1563 1134 1322
## Aug 2003  1284  877 1156
## Sep 2003  1181  793 1160
## Oct 2003  1206  732 1271
## Nov 2003  1789 1075 1455
## Dec 2003  2763 1965 2000
## Jan 2004  4310 2572 1945
## Feb 2004  7879 3350 2474
## Mar 2004  6404 3675 2299
## Apr 2004  5154 2593 1878
## May 2004  3227 2029 1755
## Jun 2004  2399 1906 1784
## Jul 2004  1989 1425 1438
## Aug 2004  1568 1045 1121
## Sep 2004  1247  786  935
## Oct 2004  1454 1139 1385
## Nov 2004  1776 1170 1640
## Dec 2004  3195 3018 1932
## Jan 2005  5829 4298 4184
## Feb 2005  4713 3821 3047
## Mar 2005  5832 3209 2433
## Apr 2005  3489 2101 1876
## May 2005  2609 2036 1746
## Jun 2005  2111 1802 1756
## Jul 2005  1686 1354 1441
## Aug 2005  1342 1026 1201
## Sep 2005  1278 1057 1224
## Oct 2005  1138  859 1402
## Nov 2005  2188 1394 1325
## Dec 2005  6433 2703 2648
## Jan 2006  4795 2762 2493
## Feb 2006  4636 3149 3498
## Mar 2006  6650 3184 2970
## Apr 2006  6195 2196 2371
## May 2006  3497 1430 1341
## Jun 2006  2679 1142 1080
## Jul 2006  2310  975 1207
## Aug 2006  2165  850 1115
## Sep 2006  2001  823  953
## Oct 2006  2766 1384 1230
## Nov 2006  3235 1297 1071
## Dec 2006  6398 2908 2327
## Jan 2007  8827 7144 5205
## Feb 2007  9086 5795 4742
## Mar 2007  5189 2905 2563
## Apr 2007  3540 2184 1434
## May 2007  2696 1732 1626
## Jun 2007  2216 1557 1395
## Jul 2007  1911 1328 1431
## Aug 2007  1536 1155 1278
```

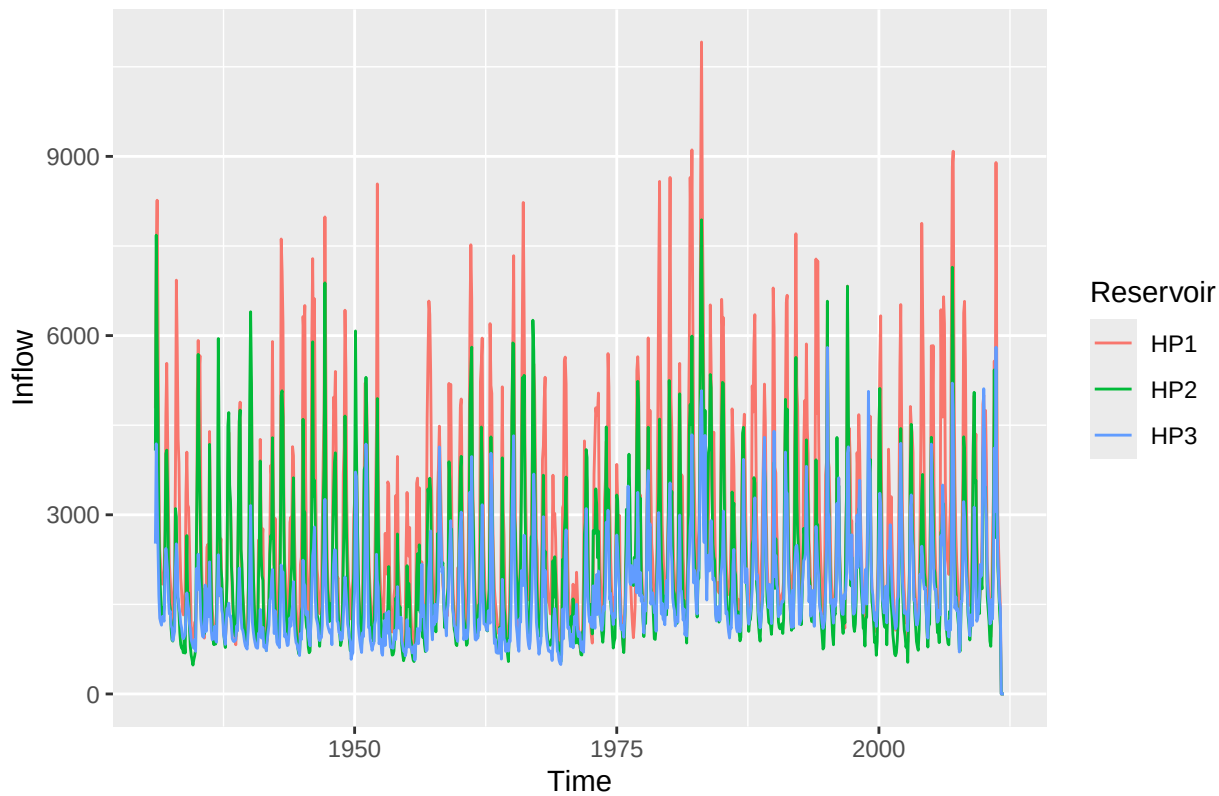
```
## Sep 2007  1240  742  703
## Oct 2007  1319  718  848
## Nov 2007  1716 1446 1550
## Dec 2007  2707 1726 1546
## Jan 2008  3611 2390 2323
## Feb 2008  6380 4305 3220
## Mar 2008  6573 4117 2995
## Apr 2008  5573 3367 2333
## May 2008  3352 2282 1970
## Jun 2008  2518 1754 1797
## Jul 2008  1991 1316 1384
## Aug 2008  1669 1156 1244
## Sep 2008  1483  904  974
## Oct 2008  1606 1104 1323
## Nov 2008  1783 1531 1265
## Dec 2008  3223 2429 1146
## Jan 2009  4263 4130 1927
## Feb 2009  4994 5050 3127
## Mar 2009  3963 3537 2415
## Apr 2009  4350 3576 1864
## May 2009  2703 2132 1217
## Jun 2009  2159 1746 1273
## Jul 2009  1831 1570 1384
## Aug 2009  1486 1345 1631
## Sep 2009  1760 1875 2167
## Oct 2009  1865 1882 1970
## Nov 2009  2613 1761 2086
## Dec 2009  4850 3826 4130
## Jan 2010  5003 4453 5110
## Feb 2010  3736 3463 4173
## Mar 2010  4747 3464 3186
## Apr 2010  3846 2488 2907
## May 2010  2197 1628 1889
## Jun 2010  1920 1423 1666
## Jul 2010  1539 1214 1481
## Aug 2010  1289  886 1173
## Sep 2010   953  798 1189
## Oct 2010  1411 1265 1580
## Nov 2010  2608 1681 1255
## Dec 2010  3338 2608 1921
## Jan 2011  5574 5427 4117
## Feb 2011  3567 2619 3061
## Mar 2011  8897 5426 5805
## Apr 2011  4991 3207 3323
## May 2011  3025 2156 2274
## Jun 2011  2415 1813 1936
## Jul 2011  1883 1426 1560
## Aug 2011  1444 1139 1441
## Sep 2011     0    0    0
## Oct 2011     0    0    0
## Nov 2011     0    0    0
## Dec 2011     0    0    0
```

Note that ts_inflow_data has information on start, end and frequency. We will discuss frequency in future

lectures, for now let's just keep the default value for frequency which is 1.

Let's see the plots generated with *autoplot()*.

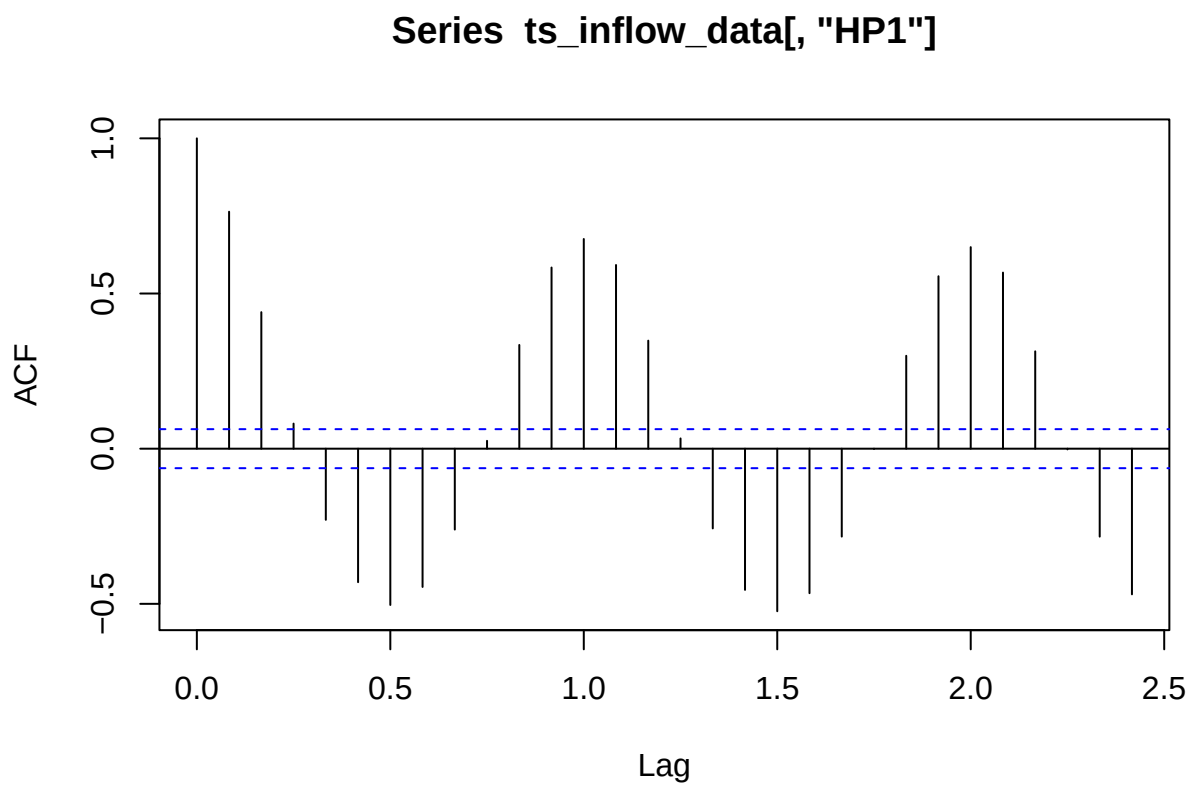
```
autoplot(ts_inflow_data) +
  xlab("Time") +
  ylab("Inflow") +
  labs(color="Reservoir")
```



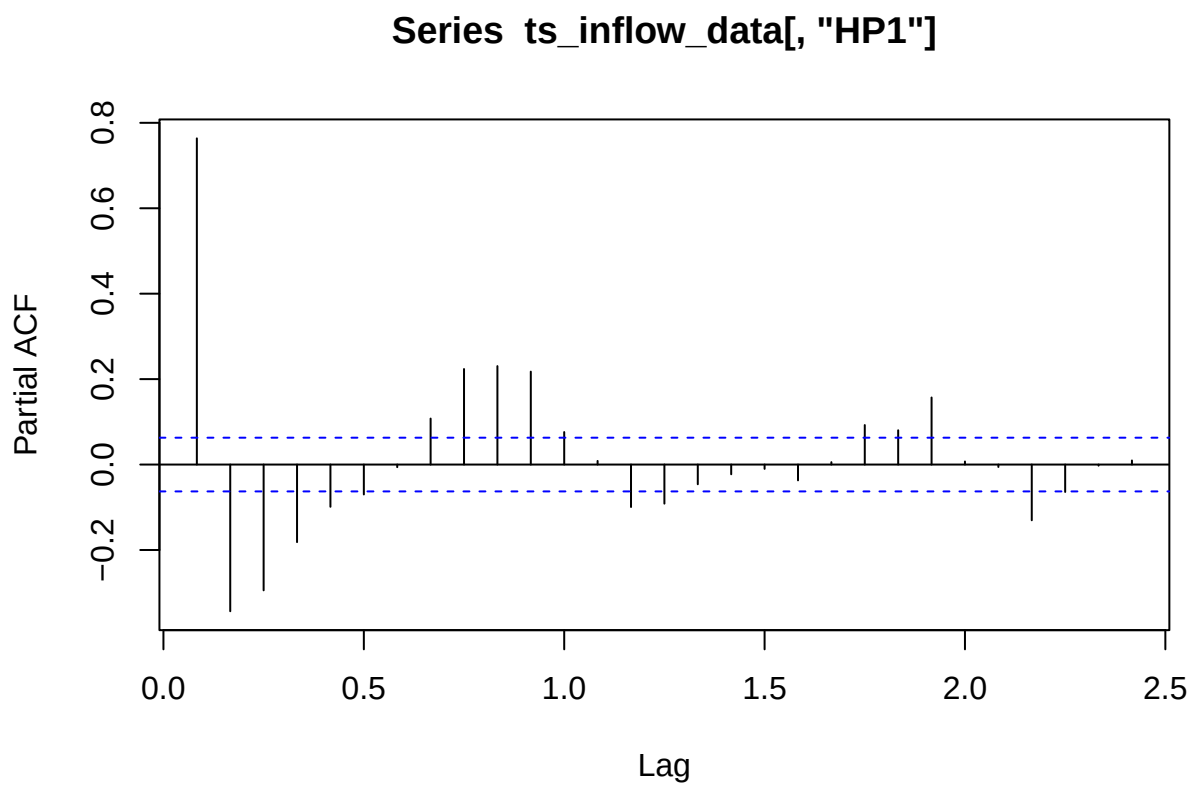
Plotting ACF and PACF

There are multiple ways to get acf and pacf plots in R. The default package "stats" have the functions acf() and pacf(). Another option is with the functions Acf() and Pacf() from package "forecast". Package forecast is not a default package. So you need to install and load it on the beginning of your scripts.

```
# Option 1: The function acf() and pacf() is from package "stats"
HP1_acf <- acf(ts_inflow_data[, "HP1"])
```

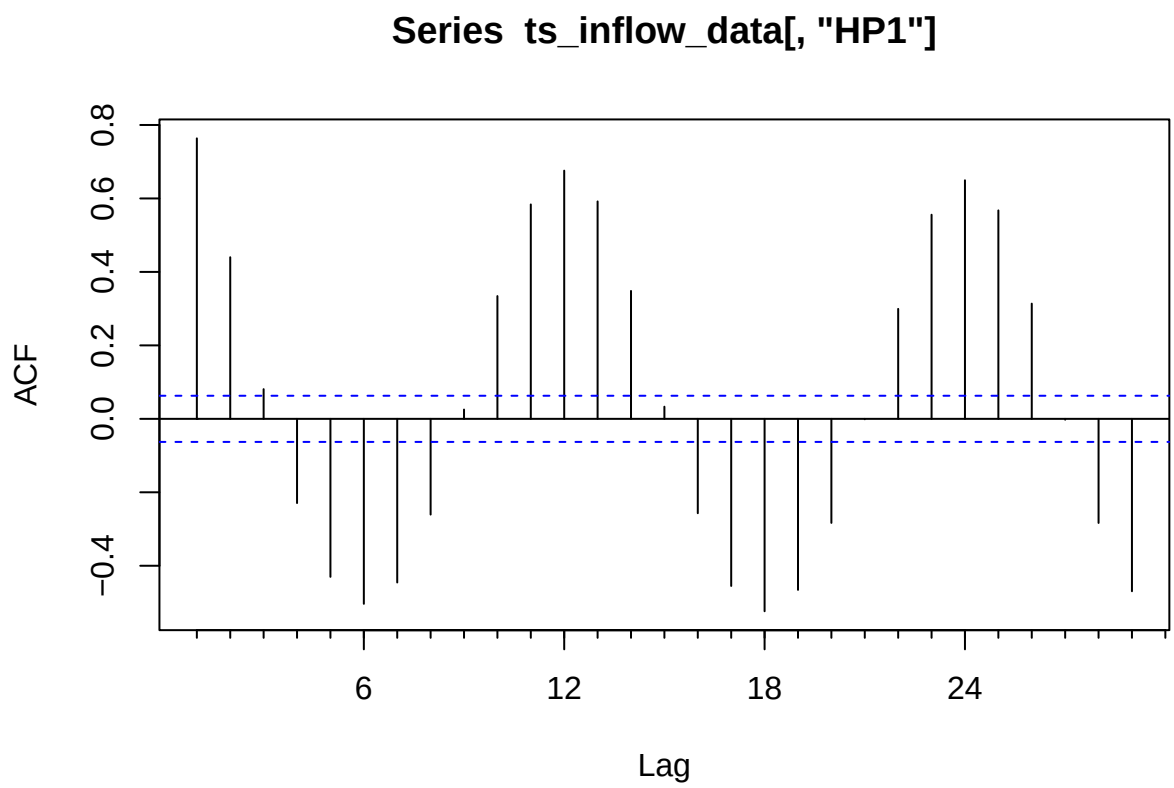



```
HP1_pacf <- pacf(ts_inflow_data[, "HP1"])
```

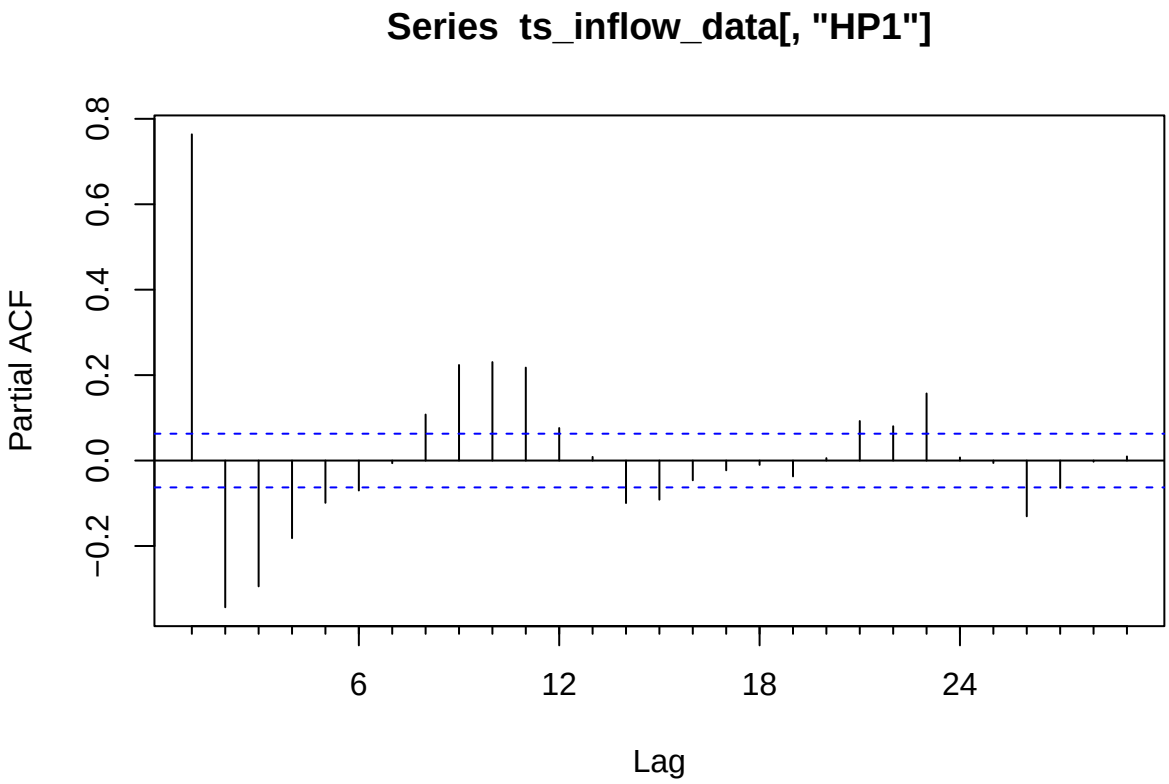



```
#Option 2: Computing ACF and PACF with package "forecast"

#The next line will output a plot since we have plot=TRUE and also store the acf values in the object
HP1_acf <- Acf(ts_inflow_data[, "HP1"], plot=TRUE)
```




```
HP1_pacf <- Pacf(ts_inflow_data[, "HP1"], plot=TRUE)
```


Series ts_inflow_data[, "HP1"]
Partial ACF
0.8
0.6
0.4
0.2
0.0
−0.2
6
12
18
24
Lag